



## **LAB ASSIGNMENT -3**

### **Operating system assignment 3**

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1. Write a shell program to add two numbers.

```
GNU nano 8.7.1 echo "Enter
first number: " read num1
echo "Enter second number: "
read num2 sum=$((num1 +
num2)) echo "Sum =$sum"
```

```
Harvinder@Ubuntu:~$ nano 1.sh
Harvinder@Ubuntu:~$ bash 1.sh
Enter first number:
12
Enter second number:
13
Sum =25
Harvinder@Ubuntu:~$
```

2. Write a shell program to compare two strings using command prompt.

```
GNU nano 8.7.1 echo "Enter
first string: " read str1
echo "Enter second string: "
read str2

if [ "$str1" = "$str2" ]
then
echo "Both strings are equal."
else
echo "Both strings are not equal."
fi
```

```
Harvinder@Ubuntu:~$ nano 2.sh
Harvinder@Ubuntu:~$ bash 2.sh
Enter first string:
harvinder
Enter second string:
singh
Both strings are not equal.
Harvinder@Ubuntu:~$
```

3. Write a shell program to generate Fibonacci series.

```

GNU nano 8.7.1 echo "Enter
number of terms: " read n a=0
b=1 echo "Fibonacci series: "
for((i=0; i<n; i++)) do echo
-n "$a " c=$((a + b)) a=$b
b=$c done echo
Harvinder@Ubuntu:~$ nano 3.sh
Harvinder@Ubuntu:~$ bash 3.sh
Enter number of terms: 10
Fibonacci series: 0 1 1 2 3 5
8 13 21 34 Harvinder@Ubuntu:~$

```

4. Write a shell program to check whether a character is VOWEL or CONSONANT using switch.

```

GNU nano 8.7.1 echo "Enter
a character: " read ch

case $ch in
    [a e i o u A E I O U])
        echo "$ch is a vowel"
        ;;
    [a-zA-Z]) echo "$ch is a
consonant" ;;
    *) echo "invalid input. Please enter a
character"
;;
esac

Harvinder@Ubuntu:~$ nano 4.sh
Harvinder@Ubuntu:~$ bash 4.sh
Enter a character: a a is a
vowel Harvinder@Ubuntu:~$

```

5. Write a shell program to display the total number of words and and total number of lines in a file.

```

GNU nano 8.7.1 echo "Enter
the filename: " read
filename

if [ -f "$filename" ] then
line=$( wc -l < "$filename" )
words=$( wc -w < "$filename" )

echo "Total no. of lines: $line"
echo "Total no. of words: $words"
else echo "File doesn't exist" fi

Harvinder@Ubuntu:~$ nano 5.sh
Harvinder@Ubuntu:~$ bash 5.sh
Enter the filename: file1.txt
Total no. of lines: 1 Total
no. of words: 2
Harvinder@Ubuntu:~$

```

6. A script that takes a filename as input and displays information about the file, such as its size, permissions, owner, and modification time.

```

GNU nano 8.7.1 echo "Enter
the filename: " read
filename

if [ -e "$filename" ]
then echo "File
information" echo
"-----" echo "File name:
$filename" echo "File size: $(stat -c '%s'
"$filename")" echo "File permission: $(stat
-c '%A' "$filename")"
echo "File owner: $(stat -c '%u' "$filename")"
echo "File modified: $(stat -c
'%y' "$filename")" else

echo "File doesn't exist"
fi

Harvinder@Ubuntu:~$ nano 6.sh
Harvinder@Ubuntu:~$ bash 6.sh
Enter the filename:
file1.txt File information
-----File name:
file1.txt File size: 13
File permission:
-rw-rw-r--Owner: Harvinder

Last Modified: 2026-09-01 06:33:49.190987109 +0000
Harvinder@Ubuntu:~$

```

7. Create a script that displays information about the system, such as the OS version, CPU, memory, disk usage, and network details.

```
GNU nano 8.7.1 echo "--- Operating
System ---" uname -a cat /etc/os-release
| grep "PRETTY_NAME"
```

```
echo -e "\n--- CPU Details ---"
lscpu | grep "Model name" lscpu
| grep "CPU(s):"
```

```
echo -e "\n--- Memory Usage ---"
free -h
```

```
echo -e "\n--- Disk Usage ---"
df -h /
```

```
echo -e "\n--- Network ---"
hostname hostname -I
```

```
Harvinder@Ubuntu:~$ nano 7.sh
Harvinder@Ubuntu:~$ bash 7.sh
--- Operating System ---
Linux Ubuntu 6.20.0-29-generic #29-Ubuntu SMP PREEMPT_DYNAMIC Fri Jul 17 20:52:35 UTC 2026 x86_64 GNU/Linux
PRETTY_NAME="Ubuntu 26.04 LTS"
```

```
--- CPU Details
---Model name:                11th Gen Intel(R) Core(TM) i5-11260H @ 2.60GHz
CPU(s):                        6
```

```
--- Memory Usage ---
MemTotal:    4.2Gi    0B
MemFree:      1.3Gi    1.6Gi
MemAvailable: 0B      0B
Swap:         0B      0B
```

```
--- Disk Usage ---
Filesystem      Size  Used Avail Use% Mounted on
/dev/sda2        40G   7.1G   30G   20% /
```

```
--- Network
---Ubuntu
10.0.2.15 fd17:625c:f037:2:a00:27ff:fe93:3f5
Harvinder@Ubuntu:~$
```